



Factory-to-Customer Shipping Route Efficiency Dashboard

Nassau Candy Distributor

Dataset Shape : (10194, 26)

Columns

- ▼ [
- 0 : "Row ID"

1 : "Order ID"

2 : "Order Date"

3 : "Ship Date"

4 : "Ship Mode"

5 : "Customer ID"

6 : "Country/Region"

7 : "City"

8 : "State/Province"

9 : "Postal Code"

10 : "Division"

11 : "Region"

12 : "Product ID"

13 : "Product Name"

14 : "Sales"

15 : "Units"

16 : "Gross Profit"

17 : "Cost"

18 : "Original_Lead_Time"

19 : "Lead Time"

20 : "Factory"

21 : "Route_State"



Dashboard Filters

Select Date Range

2024/01/02 – 2025/12/31

Select Region

All ▼

Select State

All ▼

Select Ship Mode

First Class × Same Day ×

Second Class × × ▼

Standard Class ×

Maximum Lead Time (Days)

1642



Filter Summary

Records: 10,194

Average Lead Time: 1320.84 Days

States: 59

Routes: 196

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22 : "Route_Region"
23 : "Shipping_Category"
24 : "Recommended_Shipping"
25 : "Delayed"
]
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Country/Region	City	State/Province
0	1	US-2021-103800-CHO-MIL-31000	2024-01-03 00:00:00	2026-06-30 00:00:00	Standard Class	103800	United States	Houston	Texas
1	2	US-2021-112326-CHO-TRI-54000	2024-01-04 00:00:00	2026-07-01 00:00:00	Standard Class	112326	United States	Naperville	Illinois
2	3	US-2021-112326-CHO-NUT-13000	2024-01-04 00:00:00	2026-07-01 00:00:00	Standard Class	112326	United States	Naperville	Illinois
3	4	US-2021-112326-CHO-SCR-58000	2024-01-04 00:00:00	2026-07-01 00:00:00	Standard Class	112326	United States	Naperville	Illinois
4	5	US-2021-141817-CHO-TRI-54000	2024-01-05 00:00:00	2026-07-05 00:00:00	Standard Class	141817	United States	Philadelphia	Pennsylvania

 Dashboard Overview

 Total Orders

10,194

 Average Lead Time

1320.84 Days

 Average Cost

\$4.74

 Average Gross Profit

\$9.17

 Total Routes

196

 States Covered

59

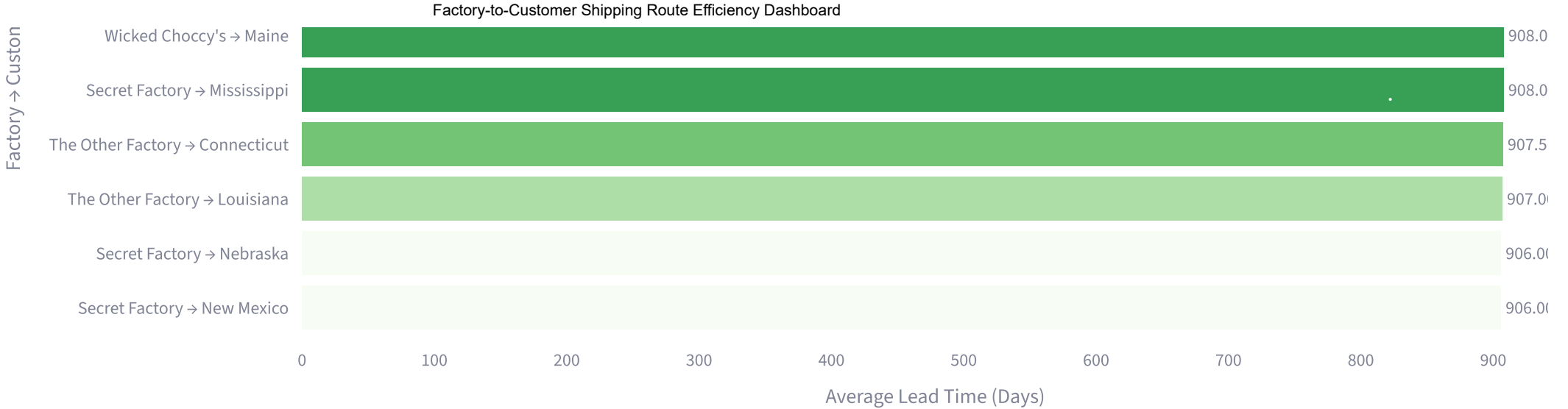
 Delay Percentage

100.0%

 Route Performance Analysis

Top 10 Fastest Factory-to-Customer Routes





 Route Performance Summary

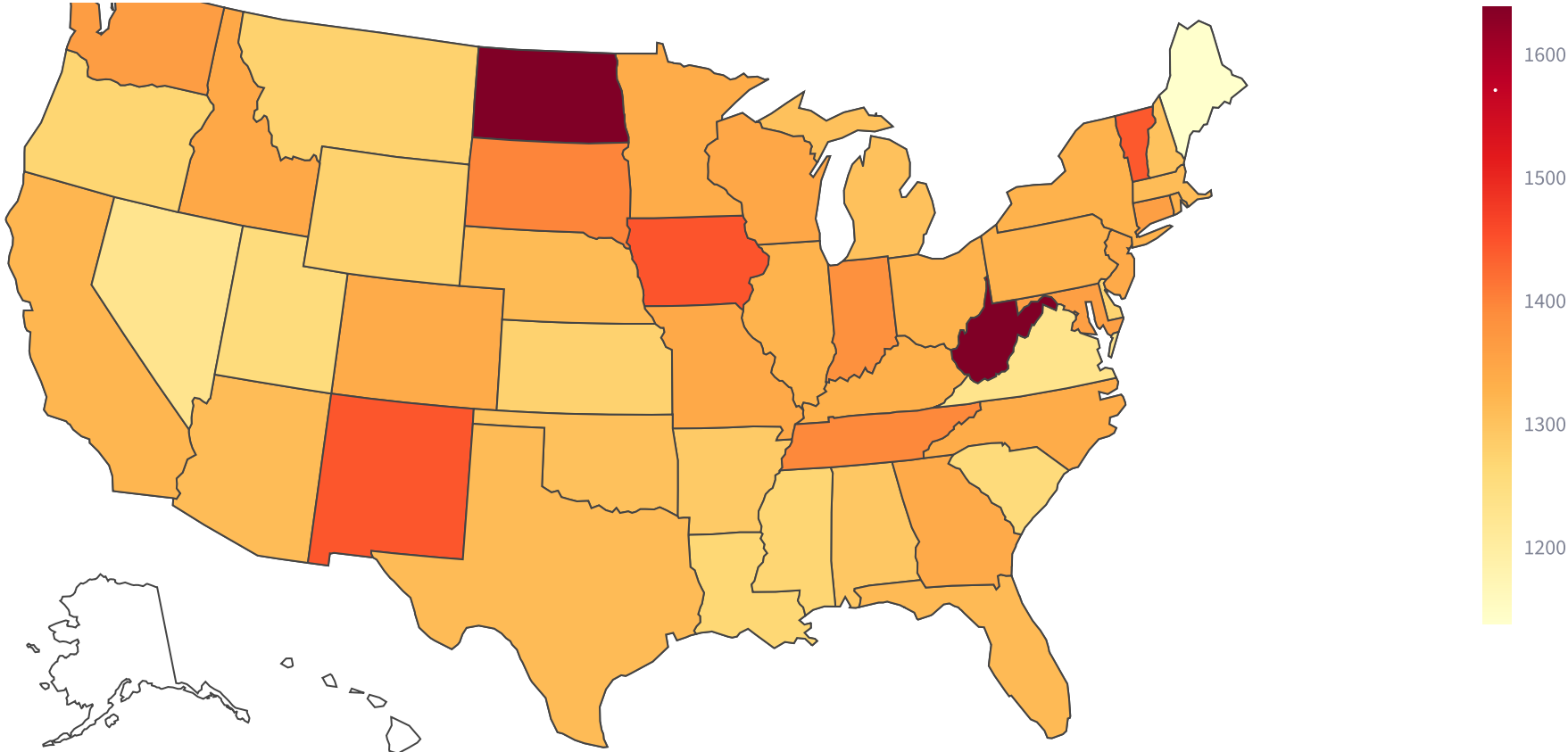
Route_State	Total_Shipments	Avg_Lead_Time	Lead_Time_Variability
Secret Factory -> New Mexico	2	906	2.83
Secret Factory -> Nebraska	1	906	0
The Other Factory -> Louisiana	1	907	0
The Other Factory -> Connecticut	2	907.5	2.12
Secret Factory -> Mississippi	1	908	0
Wicked Choccy's -> Maine	2	908	0
Secret Factory -> Louisiana	2	908.5	0.71
Secret Factory -> Delaware	1	909	0
Secret Factory -> South Carolina	1	909	0
Secret Factory -> Minnesota	1	909	0

 Geographic Shipping Performance

Average Shipping Lead Time Across U.S. States



Avg_Lead_Time



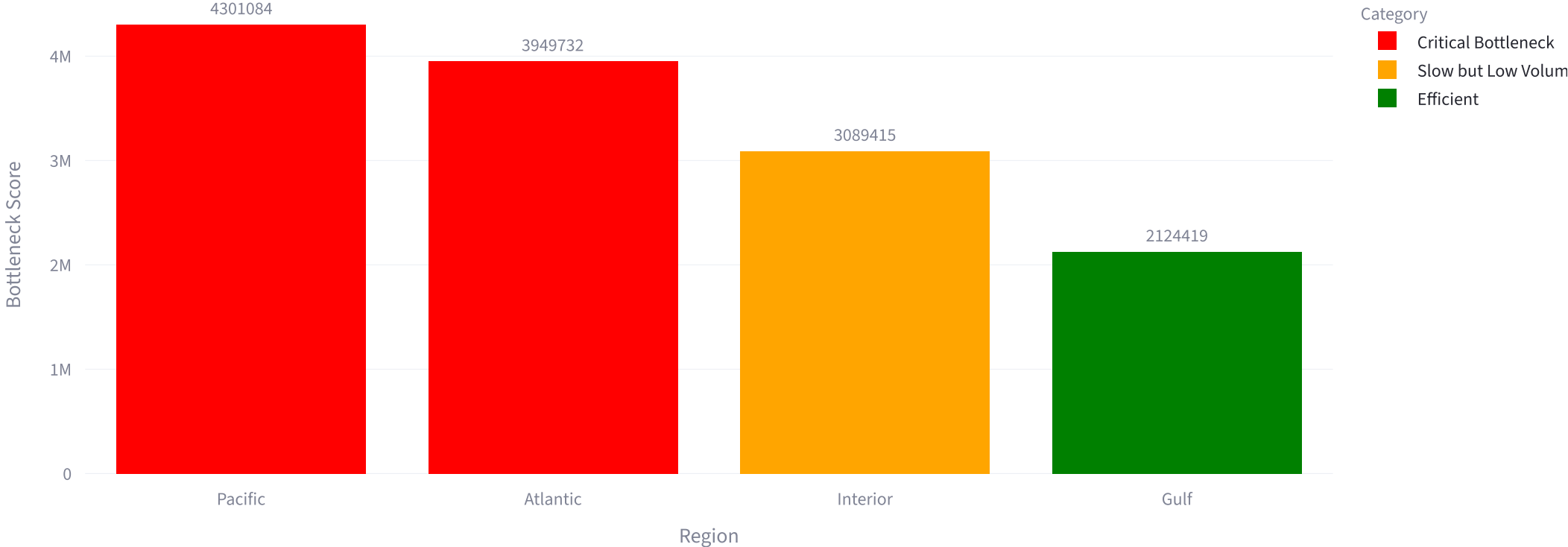
 State Shipping Performance

State/Province	Total_Shipments	Avg_Lead_Time	State_Code
West Virginia	4	1638	WV
North Dakota	7	1637.86	ND
Saskatchewan	2	1457	None
Manitoba	12	1455.33	None
Iowa	30	1443.9	IA
New Mexico	37	1441.84	NM
Vermont	11	1438.91	VT
Prince Edward Island	10	1420.3	None
South Dakota	12	1395.92	SD
Tennessee	183	1391.49	TN



Regional Bottleneck Analysis

Regional Bottleneck Score





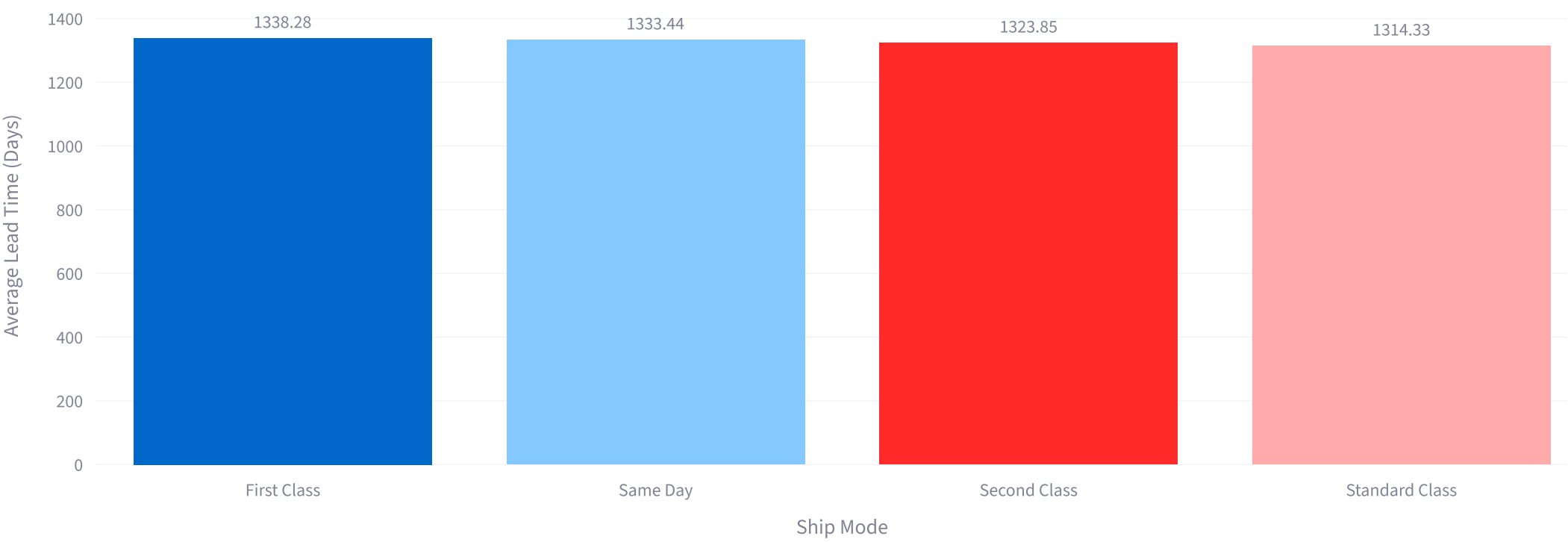
Regional Performance Summary

Region	Total_Shipments	Avg_Lead_Time	Lead_Time_Variability	Bottleneck_Score	Category
Pacific	3253	1322.19	266.47	4301084	Critical Bottleneck
Atlantic	2986	1322.75	256.54	3949732	Critical Bottleneck
Interior	2335	1323.09	262.69	3089415	Slow but Low Volume
Gulf	1620	1311.37	264.73	2124419	Efficient

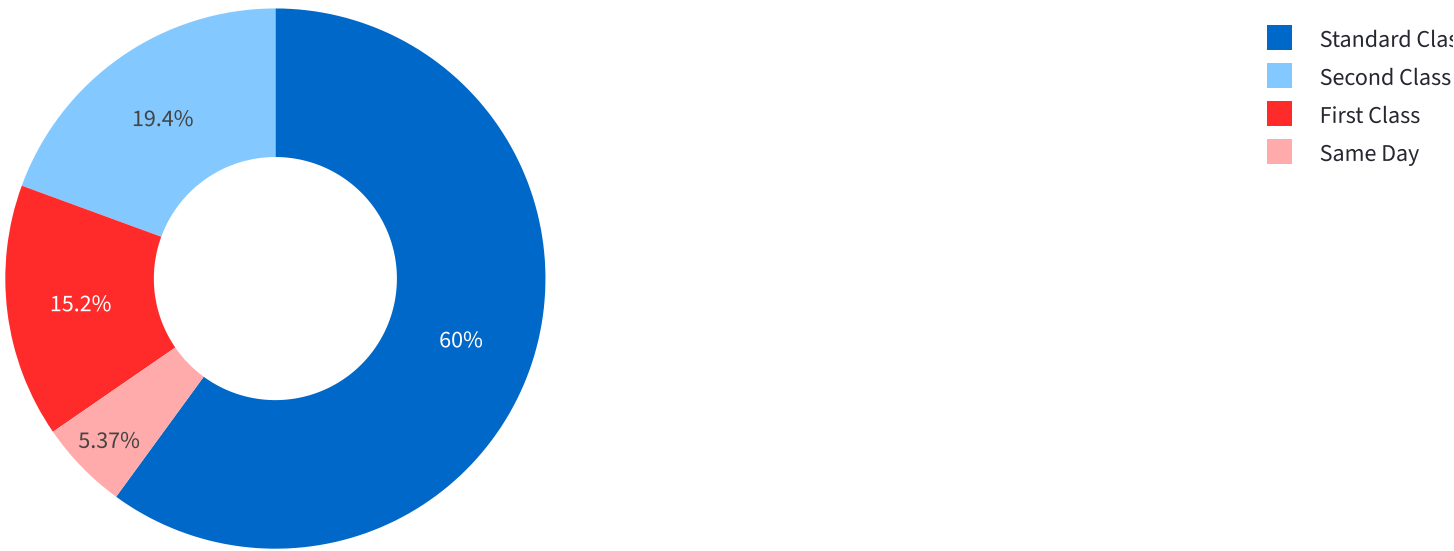


Ship Mode Performance Comparison

Average Shipping Lead Time by Ship Mode



Shipment Distribution by Ship Mode



Ship Mode Performance Summary

Ship Mode	Total_Shipments	Avg_Lead_Time	Avg_Cost	Avg_Gross_Profit	Lead_Time_Variability	Efficiency_per_Cost
Standard Class	6120	1314.33	4.75	9.22	262.4	276.701
Second Class	1979	1323.85	4.83	9.25	261.81	274.089
Same Day	547	1333.44	4.41	8.59	253.81	302.367
First Class	1548	1338.28	4.72	9.05	265.63	283.534

Best Performing Ship Mode: Standard Class

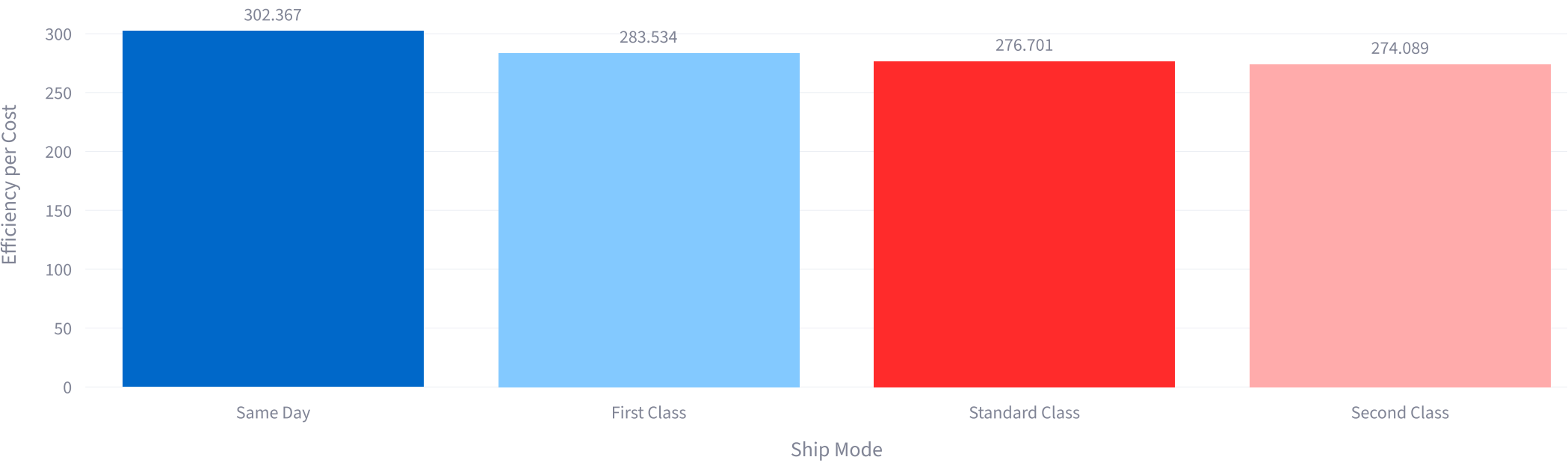
- Average Lead Time : 1314.33 Days
- Average Cost : \$4.75
- Total Shipments : 6120

Efficiency vs Cost Analysis

Efficiency vs Cost by Ship Mode



Efficiency per Cost Ranking



Efficiency vs Cost Summary

Ship Mode	Avg_Lead_Time	Avg_Cost	Total_Shipments	Efficiency_per_Cost
Same Day	1333.44	4.41	547	302.367
First Class	1338.28	4.72	1548	283.534
Standard Class	1314.33	4.75	6120	276.701
Second Class	1323.85	4.83	1979	274.089



Best Cost-Efficient Shipping Mode

Ship Mode: Same Day

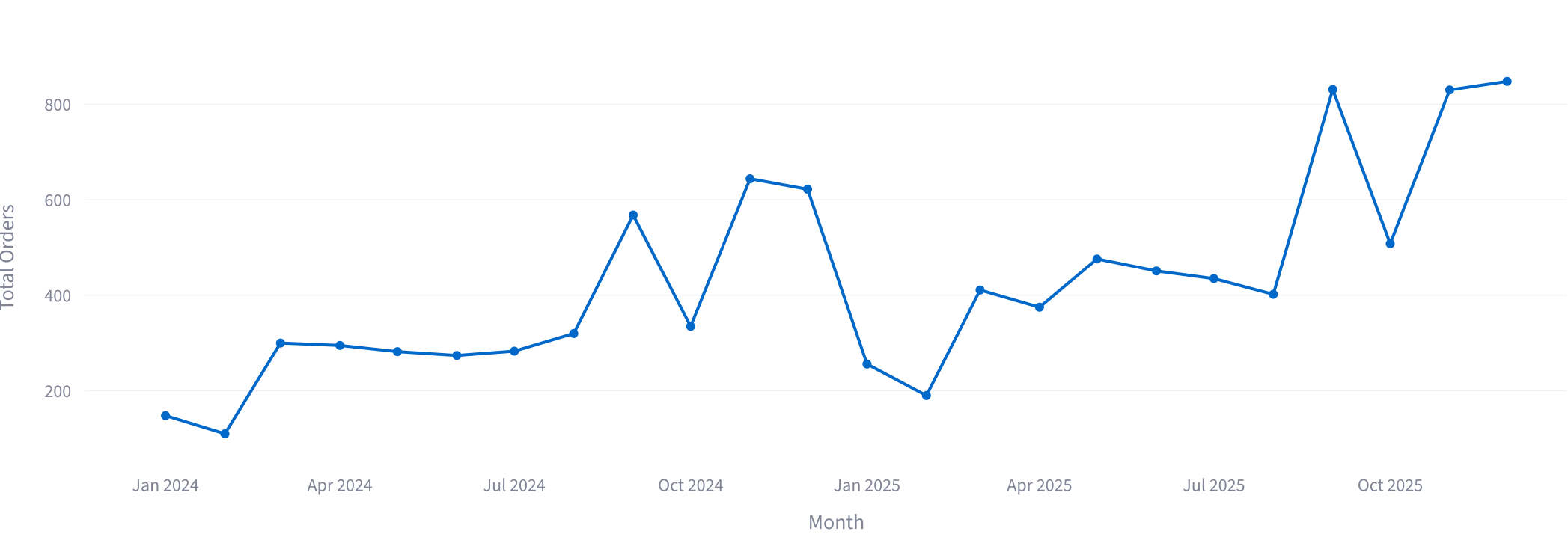
- Efficiency per Cost : 302.367
- Average Lead Time : 1333.44 Days
- Average Cost : \$4.41
- Total Shipments : 547

This shipping mode provides the highest operational efficiency relative to its average transportation cost.

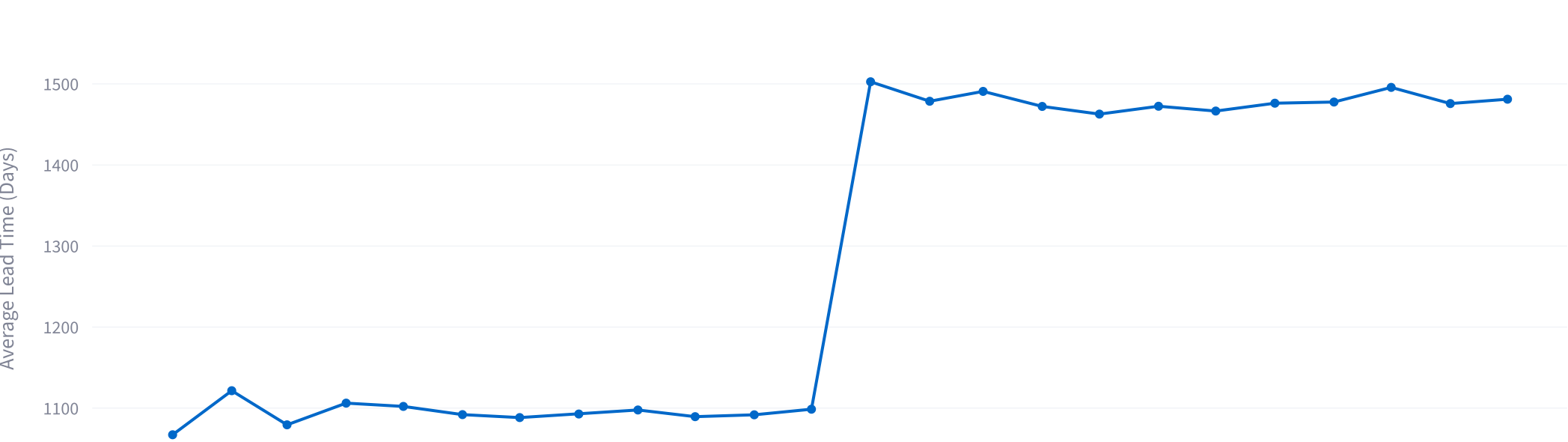


Shipping Timeline Analysis

Monthly Order Volume



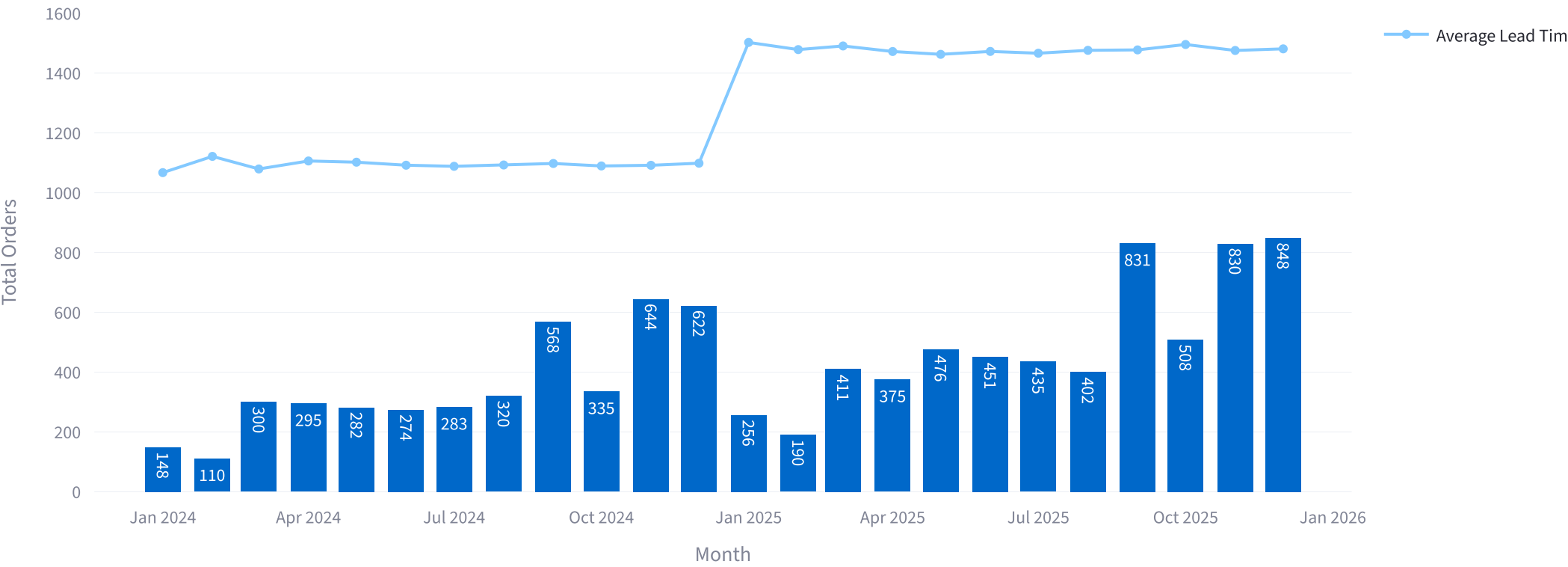
Monthly Average Shipping Lead Time



Jan 2024 Apr 2024 Jul 2024 Oct 2024 Jan 2025 Apr 2025 Jul 2025 Oct 2025

Month

Orders and Average Lead Time by Month



Monthly Shipping Performance

Month	Total_Orders	Avg_Lead_Time	Avg_Cost	Avg_Gross_Profit
2024-01	148	1067.16	4.92	9.23
2024-02	110	1121.5	4.28	8.26
2024-03	300	1079.42	4.89	8.9
2024-04	295	1106.17	4.66	8.86
2024-05	282	1102.12	4.74	9.27
2024-06	274	1091.99	4.34	8.51
2024-07	283	1088.35	4.59	9.06
2024-08	320	1092.75	4.79	9.21
2024-09	568	1097.73	4.76	9.17
2024-10	335	1089.47	5.06	9.31



Timeline Insights

- Highest shipment volume occurred in 2025-12 with 848 orders.
- Fastest average delivery performance was achieved in 2024-01 with an average lead time of 1067.16 days.



Route Drill-Down Analysis

Select a Factory-to-Customer route to explore detailed shipment performance.

Select Route

Lot's O' Nuts → Alabama

▼

Shipments

34

Avg Lead Time

1295.21 Days

Avg Cost

\$4.37

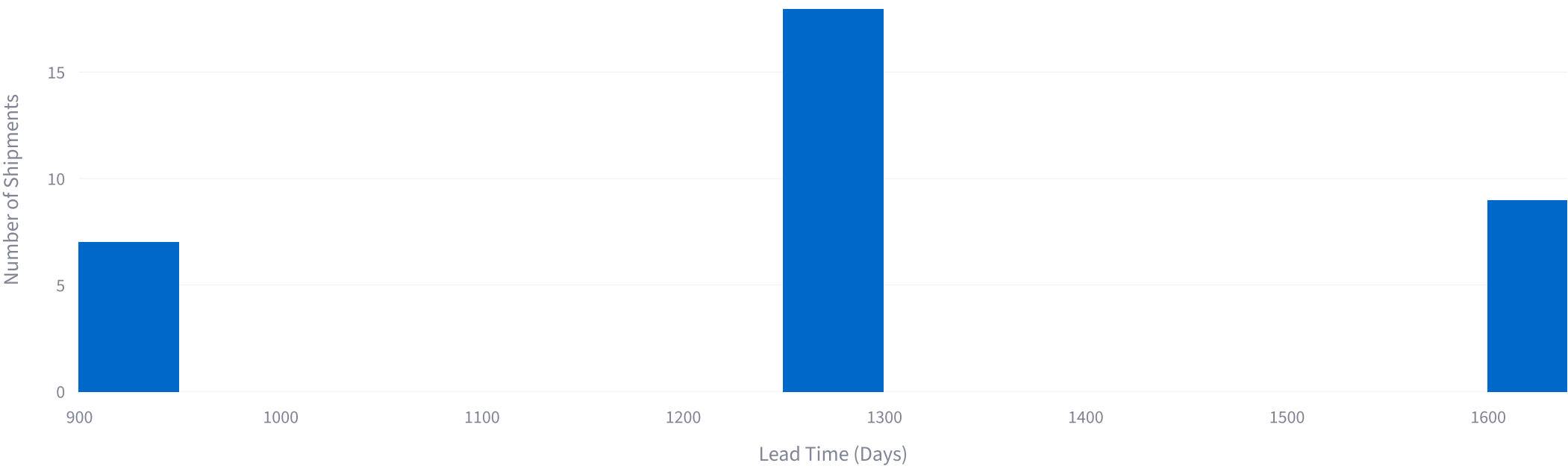
Avg Profit

\$9.62

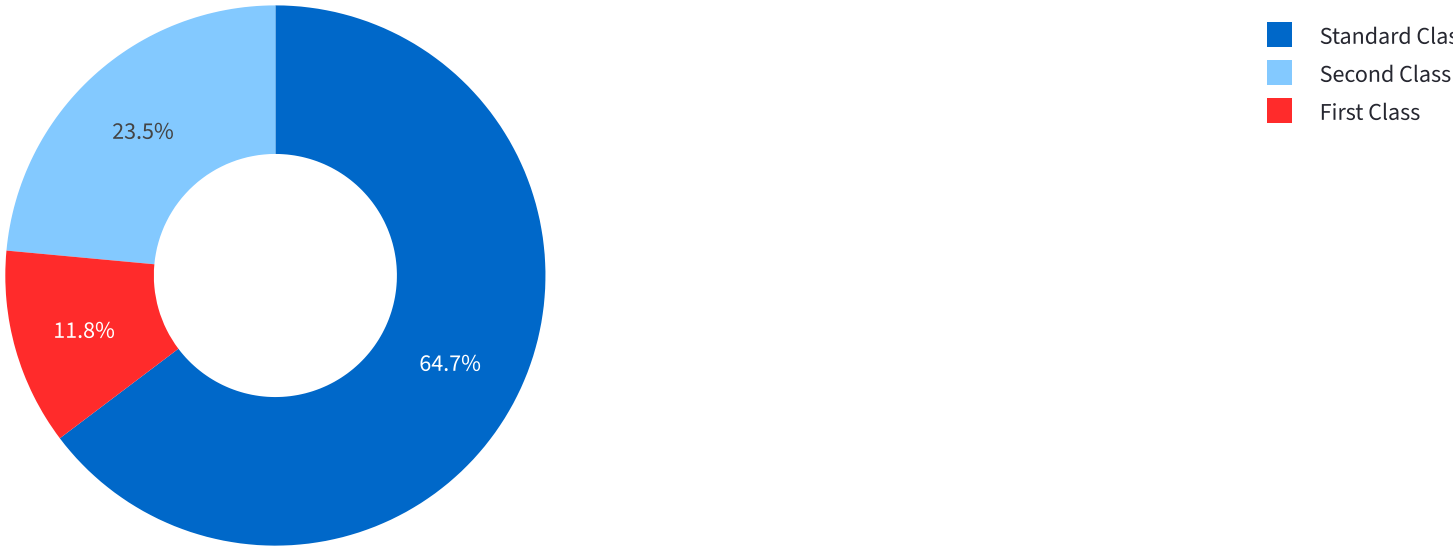
Lead Time Std

253.07

Lead Time Distribution
Lot's O' Nuts → Alabama



Ship Mode Distribution



Shipment Details

Order ID	Order Date	Ship Date	Ship Mode	State/Province	Cost	Gross Profit	Lead Time
US-2022-134257-CHO-NUT-13000	2024-03-16 00:00:00	2027-09-09 00:00:00	Second Class	Alabama	2	4.98	1272
US-2022-134257-CHO-FUD-51000	2024-03-16 00:00:00	2027-09-09 00:00:00	Second Class	Alabama	6	12	1272
US-2022-121552-CHO-NUT-13000	2024-03-22 00:00:00	2027-09-17 00:00:00	Standard Class	Alabama	4	9.96	1274
US-2021-118997-CHO-FUD-51000	2024-04-08 00:00:00	2026-10-03 00:00:00	Standard Class	Alabama	9.6	19.2	908
US-2021-143840-CHO-FUD-51000	2024-05-22 00:00:00	2026-11-19 00:00:00	Standard Class	Alabama	2.4	4.8	911
US-2021-143840-CHO-NUT-13000	2024-05-22 00:00:00	2026-11-19 00:00:00	Standard Class	Alabama	1	2.49	911
US-2022-104941-CHO-SCR-58000	2024-06-13 00:00:00	2027-12-10 00:00:00	Standard Class	Alabama	3.3	7.5	1275
US-2022-104941-CHO-SCR-58000	2024-06-13 00:00:00	2027-12-10 00:00:00	Standard Class	Alabama	3.3	7.5	1275
US-2022-104941-CHO-SCR-58000	2024-06-13 00:00:00	2027-12-10 00:00:00	Standard Class	Alabama	5.5	12.5	1275
US-2022-154291-CHO-FUD-51000	2024-07-25 00:00:00	2028-01-19 00:00:00	Standard Class	Alabama	10.8	21.6	1273

Route Summary

Selected Route: Lot's O' Nuts → Alabama

- Total Shipments: **34**
- Average Lead Time: **1295.21 Days**
- Average Transportation Cost: **\$4.37**
- Average Gross Profit: **\$9.62**
- Lead Time Variability: **253.07 Days**

This section enables detailed operational monitoring of individual factory-to-customer routes and helps identify high-performing as well as underperforming logistics routes.



Download Filtered Data

Download the filtered shipment data for further analysis in Excel, Power BI, or other reporting tools.



Download Filtered Dataset (CSV)



Filtered Dataset Summary

Metric	Value
Total Records	10194
Total Routes	196
States Covered	59
Average Lead Time (Days)	1320.84
Average Cost (\$)	4.74
Average Gross Profit (\$)	9.17

The downloaded CSV contains only the records matching the selected dashboard filters.

Dashboard Notes

- Order-to-Shipment Duration is calculated directly from the Order Date and Ship Date available in the dataset.
- Dashboard filters update all KPIs, charts and tables dynamically.
- Developed using Python, Streamlit and Plotly.

Project: Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor